

Many-Body Green's Functions & Quasi-particles

March 3, 2025

Outline

- ▶ Introduction: Coupled Systems and Reduced Hilbert Space
- ▶ Section 7.1: Dynamical Equations and Green's Functions
- ▶ Section 7.2: The Self-Energy and the Dyson Equation
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- ▶ Section 7.4: Interacting Systems, Self-Energy and Spectral Function
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- ▶ Section 7.7: Separating Contributions to Dyson Equations
- ▶ Summary and Outlook

Introduction to Coupled Systems

- ▶ In many-body physics, we often focus on a *small* subspace (S) containing the most relevant degrees of freedom.
- ▶ The rest of the Hilbert space (R) is integrated out; its effect appears through a frequency-dependent self-energy.
- ▶ This approach simplifies the full many-body problem while retaining key correlation effects.

Section 7.1: The Small Space Concept

- ▶ The full many-body Hamiltonian acts in an enormous Hilbert space (exponential in the number of electrons).
- ▶ We choose a tractable *small space* S (a set of relevant many-body states) and treat the remainder R approximately.
- ▶ All effects of R appear via a frequency-dependent self-energy.

Block Structure and Eigenvalue Problem

Consider the full eigenvalue problem in the many-body Hilbert space:

$$\begin{pmatrix} H_S & H_{SR} \\ H_{RS} & H_R \end{pmatrix} \begin{pmatrix} \phi_S \\ \phi_R \end{pmatrix} = \omega \begin{pmatrix} \phi_S \\ \phi_R \end{pmatrix}. \quad (7.1)$$

- ▶ H_S : Hamiltonian in the chosen “small” space S .
- ▶ H_R : Hamiltonian in the remainder R of the Hilbert space.
- ▶ H_{SR} and H_{RS} : Coupling between S and R .
- ▶ ϕ_S and ϕ_R are the eigenvector components in the respective subspaces.
- ▶ ω is the excitation energy or frequency (with $\hbar = 1$) of the eigenvector.

Eliminating the R-Space

- ▶ Solve the full eigenvalue problem by eliminating the R component.
- ▶ This yields an effective equation in S :

$$\left[H_S - H_{SR}(H_R - \omega)^{-1}H_{RS} \right] \phi_1 = \omega \phi_1.$$

- ▶ The effect of H_R is encapsulated in the term $H_{SR}(H_R - \omega)^{-1}H_{RS}$.

Connecting the Eigenvalue Problem and the Green's Function

► Green's Function Formulation :

$$(\omega - H) G(\omega) = I,$$

where

$$G(\omega) = (\omega - H)^{-1}.$$

- The poles of $G(\omega)$ occur at the eigenvalues ω from Eq. 7.1.
- By projecting $G(\omega)$ onto the small space, one obtains the effective Green's function $G_S(\omega)$.
- The resolvent $G(\omega)$ contains the spectral information (i.e., the eigenvalues and eigenvectors) of H . Thus, solving the eigenvalue problem in Eq. 7.1 is equivalent to finding the poles of $G(\omega)$, and by focusing on $G_S(\omega)$, we capture the influence of the rest of the Hilbert space via the self-energy.

Effective Green's Function in S

- ▶ The Green's function for the small space is given by:

$$G_S(\omega) = \left[\omega - H_S - H_{SR}(\omega - H_R)^{-1}H_{RS} \right]^{-1}.$$

- ▶ This is the starting point for deriving the Dyson equation.

The Dyson Equation

- ▶ Writing the effective Green's function as:

$$G_S(\omega) = \left[[G_S^0(\omega)]^{-1} - \Sigma_S(\omega) \right]^{-1},$$

where

$$G_S^0(\omega) = (\omega - H_S)^{-1},$$

and the self-energy is defined by

$$\Sigma_S(\omega) = H_{SR}(\omega - H_R)^{-1}H_{RS}.$$

Dyson Series Expansion

- ▶ Starting from

$$G_S(\omega) = \left[[G_S^0(\omega)]^{-1} - \Sigma_S(\omega) \right]^{-1},$$

we can rewrite it as:

$$G_S(\omega) = G_S^0(\omega) [1 - G_S^0(\omega) \Sigma_S(\omega)]^{-1}.$$

- ▶ Expanding using the geometric series (if the series converges):

$$G_S(\omega) = G_S^0(\omega) + G_S^0(\omega) \Sigma_S(\omega) G_S^0(\omega) + \cdots .$$

- ▶ This is Equation 7.9.

Dyson Equation graphically

$$\begin{aligned} \overrightarrow{G} &= \overrightarrow{G^0} + \overrightarrow{G^0} \text{ (circle) } \overrightarrow{G^0} + \overrightarrow{G^0} \text{ (circle) } \overrightarrow{G^0} \text{ (circle) } \overrightarrow{G^0} + \dots \\ \overrightarrow{G} &= \overrightarrow{G^0} + \overrightarrow{G^0} \text{ (circle) } \overrightarrow{G} \end{aligned}$$

Figure 7.1. Schematic depiction of the Dyson equation. In the top line the Green's function is expressed as a sum of repeated processes (Eq. (7.9)), which is an expansion of the form in the bottom line (Eq. (7.10)). The circles represent a self-energy. The relations are not restricted to one-particle G and Σ for an electron. The form is general and applies to any Green's function and derived quantities: the screened interaction W and polarization P , response functions, etc.

Self-Energy: Definition and Role

- ▶ **Definition:** The self-energy in the small subspace is defined as:

$$\Sigma_S(\omega) = [G_S^0(\omega)]^{-1} - [G_S(\omega)]^{-1}.$$

- ▶ It encapsulates the effects of coupling the small (tractable) space S to the rest of the system R .
- ▶ $\Sigma_S(\omega)$ is frequency-dependent, reflecting that the effective interactions vary with energy.
- ▶ The term $[G_S^0(\omega)]^{-1}$ represents the bare (or mean-field) propagation within S .
- ▶ The full propagator $[G_S(\omega)]^{-1}$ includes all corrections due to interactions with R .
- ▶ Thus, $\Sigma_S(\omega)$ quantifies the many-body corrections beyond the bare picture.
- ▶ In Feynman diagrams, $\Sigma_S(\omega)$ is the sum of all irreducible diagrams—those that cannot be split into two by cutting a single bare propagator line.
- ▶ It represents all the complex processes (like screening, vertex corrections, etc.) that dress the electron propagator.
- ▶ **Role in Dyson Equation:** It acts as an effective, energy-dependent potential:

$$G_S(\omega) = \left[[G_S^0(\omega)]^{-1} - \Sigma_S(\omega) \right]^{-1}.$$

The poles of $G_S(\omega)$ (which determine quasi-particle energies and lifetimes) are shifted and broadened by $\Sigma_S(\omega)$.

Dyson Equation for Screening

- ▶ Analogously, for the screened Coulomb interaction:

$$W(\omega) = v_c + v_c P(\omega) W(\omega),$$

where $P(\omega)$ is the proper polarizability.

- ▶ In operator form:

$$W(\omega) = \left[v_c^{-1} - P(\omega) \right]^{-1}.$$

Section 7.3: A Single State Coupled to a Continuum

- ▶ For a system with a discrete state (labeled 0) and continuum states (labeled by l), the Hamiltonian is:

$$\hat{H} = \varepsilon_0 c_0^\dagger c_0 + \sum_{l=1} t_{0l} (c_l^\dagger c_0 + c_0^\dagger c_l) + \sum_{l=1} \varepsilon_l c_l^\dagger c_l.$$

$$H = \begin{pmatrix} \varepsilon_0 & t_{01} & t_{02} & \cdots \\ t_{10} & \varepsilon_1 & 0 & \cdots \\ t_{20} & 0 & \varepsilon_2 & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}.$$

- ▶ The corresponding Green's function components satisfy:
 - ▶ For the discrete state:

$$(\omega - \varepsilon_0) G_{00}(\omega) - \sum_l t_{0l} G_{l0}(\omega) = 1.$$

- ▶ For the coupling between continuum and discrete state:

$$(\omega - \varepsilon_l) G_{l0}(\omega) - t_{l0} G_{00}(\omega) = 0.$$

- ▶ For the continuum–continuum elements:

$$(\omega - \varepsilon_l) G_{ll'}(\omega) - t_{l0} G_{0l'}(\omega) = \delta_{ll'}.$$

Constructing the Green's Function Components Equations

- ▶ We start from the full matrix equation:

$$(\omega - H)G(\omega) = I.$$

- ▶ In component form, with indices i, j :

$$\sum_k \left[\omega \delta_{ik} - H_{ik} \right] G_{kj}(\omega) = \delta_{ij}.$$

- ▶ **For $i = 0, j = 0$ (discrete state):**

$$\omega G_{00} - H_{00} G_{00} - \sum_{l \neq 0} H_{0l} G_{l0} = 1,$$

$$(\omega - \varepsilon_0) G_{00} - \sum_l t_{0l} G_{l0} = 1.$$

- ▶ **For $i = l, j = 0$ (coupling discrete-continuum):**

$$\omega G_{l0} - H_{l0} G_{00} - H_{ll} G_{l0} = 0,$$

$$(\omega - \varepsilon_l) G_{l0} - t_{l0} G_{00} = 0.$$

- ▶ **For $i = l, j = l'$ (cont-cont):**

$$\omega G_{ll'} - H_{l0} G_{0l'} - H_{ll} G_{ll'} = \delta_{ll'},$$

$$(\omega - \varepsilon_l) G_{ll'} - t_{l0} G_{0l'} = \delta_{ll'}.$$

Derivation of Eq. 7.14

- Solve the continuum equation:

$$G_{I0}(\omega) = \frac{t_{I0}}{\omega - \varepsilon_I} G_{00}(\omega).$$

- Substitute back into the discrete equation:

$$(\omega - \varepsilon_0) G_{00}(\omega) - \sum_I \frac{t_{0I} t_{I0}}{\omega - \varepsilon_I} G_{00}(\omega) = 1.$$

- Factor out $G_{00}(\omega)$:

$$\left[\omega - \varepsilon_0 - \Delta(\omega) \right] G_{00}(\omega) = 1,$$

where

$$\Delta(\omega) = \sum_I \frac{t_{0I} t_{I0}}{\omega - \varepsilon_I}.$$

- Hence,

$$G_{00}(\omega) = \frac{1}{\omega - \varepsilon_0 - \Delta(\omega)}.$$

Scaling in the Continuum Limit

- ▶ In the continuum limit, the density of states increases.
- ▶ To ensure a finite contribution to $\Delta(\omega)$, each matrix element t_{0l} must scale as $1/\sqrt{N}$ (or similar).
- ▶ This normalization keeps the overall hybridization function finite.

Section 7.4: Interacting Systems

- ▶ Now consider a system with interactions:

$$\hat{H} = \hat{H}_0 + \hat{V},$$

where $\hat{V}$ involves electron–electron interactions.

- ▶ The one-electron Green's function (for electron addition) is defined as:

$$G_{ij}^{\text{el}}(\omega) = \langle N, 0 | c_i \frac{1}{\omega - H + E_N} c_j^\dagger | N, 0 \rangle.$$

(Equation 7.19)

Lehmann Representation

- By inserting a complete set of $(N + 1)$ -electron eigenstates, the Green's function can be written as:

$$G_{ij}^{\text{el}}(\omega) = \sum_m \frac{\langle N, 0 | c_i | N + 1, m \rangle \langle N + 1, m | c_j^\dagger | N, 0 \rangle}{\omega - (E_{N+1,m} - E_N) + i\eta}.$$

- In practice, we write the compact form (Eq. 7.19) without explicitly inserting the complete set.

From G_0 to G via Self-Energy

- ▶ In the non-interacting limit, we obtain:

$$G_0(\omega) = \frac{1}{\omega - \varepsilon_k^0}.$$

- ▶ In the interacting system, the full Green's function is:

$$G_k(\omega) = \frac{1}{\omega - \varepsilon_k^0 - \Sigma_k(\omega)}.$$

(Equation 7.21)

Spectral Function

- ▶ The spectral function is defined as:

$$A_k(\omega) = \frac{1}{\pi} \frac{|\operatorname{Im} \Sigma_k(\omega)|}{[\omega - \varepsilon_k^0 - \operatorname{Re} \Sigma_k(\omega)]^2 + [\operatorname{Im} \Sigma_k(\omega)]^2}.$$

- ▶ For vanishing self-energy, $A_k(\omega)$ is a δ -function; with interactions, peaks are broadened and may shift.

Spectral Function

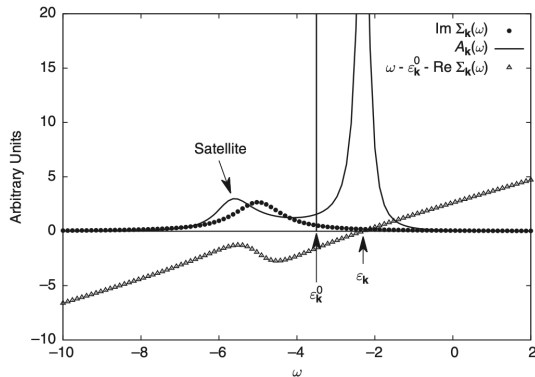


Figure 7.2. Schematic illustration of an element of the spectral function for removal of electrons below the Fermi energy ($\omega = 0$). In the independent-particle case, the spectral function is diagonal in the appropriate basis, and each diagonal element $\mathbf{k}$ is a δ -function, situated at $\epsilon_{\mathbf{k}}^0$. Through the interaction, the real part of the self-energy (empty triangles represent the function $\omega - \epsilon_{\mathbf{k}}^0 - \text{Re } \Sigma_{\mathbf{k}}(\omega)$) shifts the main peak to the quasi-particle energy $\epsilon_{\mathbf{k}}$. The imaginary part (full circles) is small but non-vanishing around $\epsilon_{\mathbf{k}}$, which leads to broadening. Still, in this example one can relate the resulting structure to the non-interacting one and interpret it as a dressed particle, i.e., a *quasi-particle*. The self-energy contains in principle all possible excitations of the system in the presence of an additional particle, which leads to structure in $\text{Im } \Sigma$. This structure shows up as satellites in the spectral function; there is one satellite in this figure, indicated by an arrow. The total

Spectral Function: HEG

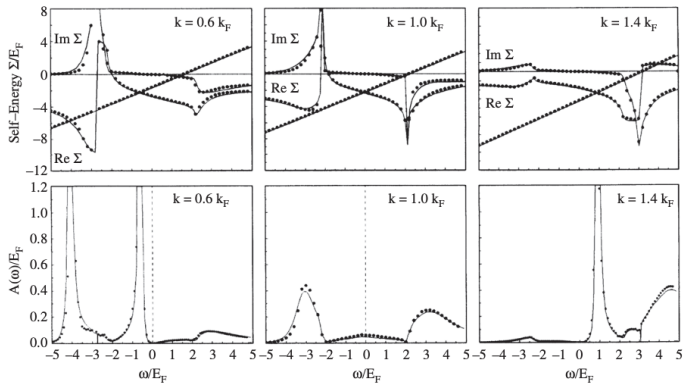


Figure 11.5. Real and imaginary parts of the self-energy of the homogeneous electron gas (upper panels) and spectral functions (lower panels) for three different momenta, from [385]. Solid lines, G_0W_0 ; dots, including a vertex correction discussed in Sec. 15.2. The diagonal line in the upper panel is the function $(\omega - \varepsilon_k)$ with ε_k the Hartree eigenvalue. The quasi-particle peaks in the spectral function appear where $\text{Re } \Sigma_{xc}(k, \omega)$ crosses this line. Here $r_s = 5$.

Spectral Function: Si

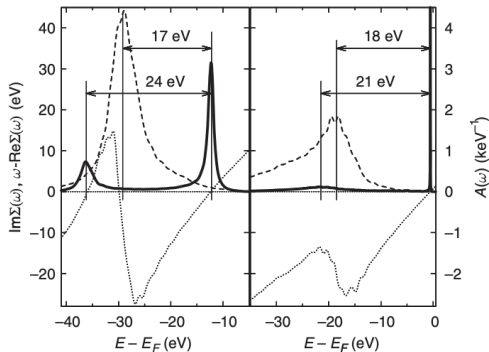


Figure 13.1. Matrix elements $A_{\ell\ell}(\omega)$ of the G_0W_0 spectral function of bulk silicon (continuous lines) at the Γ point for a state ℓ at the bottom (left panel) and top of the valence bands (right panel). The corresponding imaginary part of the matrix elements $\Sigma_{\ell\ell}(\omega)$ of the self-energy is given by the dashed curves. The dotted curve is the function $\omega - \epsilon_{\ell}^{KS} - (\text{Re}\Sigma_{xc} - v_{xc})_{\ell\ell}$, which crosses zero at the quasi-particle energy. The top of the valence band is set to 0 eV. The figure has been adapted from [521] by M. Guzzo. Note that here selected matrix elements as a function of frequency are displayed. This is different from Fig. 13.2, where each matrix element is taken at the corresponding quasi-particle energy.

Section 7.5: Quasi-particles

- ▶ Quasi-particles are the dressed excitations of an interacting system.
- ▶ Near the Fermi energy, $\text{Im } \Sigma_k(\omega) \sim (\omega - E_F)^2$, so quasi-particle peaks are sharp.
- ▶ The spectral function exhibits a quasi-particle peak and satellite features (e.g., due to plasmons).

Quasi-particle Energy and Renormalization

- ▶ The quasi-particle energy satisfies:

$$\varepsilon_k = \varepsilon_k^0 + \text{Re } \Sigma_k(\varepsilon_k).$$

- ▶ Linearizing near ε_k^0 leads to:

$$\text{Re } \Sigma_k(\omega) \approx \text{Re } \Sigma_k(\varepsilon_k^0) + (\omega - \varepsilon_k^0) \left. \frac{\partial \text{Re } \Sigma_k}{\partial \omega} \right|_{\omega=\varepsilon_k^0}.$$

- ▶ Define renormalization factor:

$$Z_k = \frac{1}{1 - \left. \frac{\partial \text{Re } \Sigma_k}{\partial \omega} \right|_{\omega=\varepsilon_k^0}}.$$

- ▶ Thus,

$$\varepsilon_k = \varepsilon_k^0 + Z_k \text{Re } \Sigma_k(\varepsilon_k^0).$$

(Eqs. 7.25–7.27)

Quasi-particle Peak in the Spectral Function

- ▶ Near the quasi-particle energy, the spectral function is approximately Lorentzian:

$$A_k(\omega) \approx \frac{Z_k}{\pi} \frac{Z_k \operatorname{Im} \Sigma_k(\varepsilon_k)}{(\omega - \varepsilon_k)^2 + [Z_k \operatorname{Im} \Sigma_k(\varepsilon_k)]^2}.$$

- ▶ Z_k is the quasi-particle weight, indicating the spectral weight retained in the coherent peak.

Effective Mass and Dispersion

- Differentiate the quasi-particle energy condition:

$$\frac{d\varepsilon_k}{dk} = \frac{d\varepsilon_k^0}{dk} + \left. \frac{\partial \operatorname{Re} \Sigma_k}{\partial k} \right|_{\omega=\varepsilon_k} + \left. \frac{\partial \operatorname{Re} \Sigma_k}{\partial \omega} \right|_{\omega=\varepsilon_k} \frac{d\varepsilon_k}{dk}.$$

- Isolate $\frac{d\varepsilon_k}{dk}$ and define Z_k :

$$\frac{d\varepsilon_k}{dk} = Z_k \left[\frac{d\varepsilon_k^0}{dk} + \left. \frac{\partial \operatorname{Re} \Sigma_k}{\partial k} \right|_{\omega=\varepsilon_k} \right].$$

- This equation shows that the dispersion (and hence the effective mass) is renormalized.

Quasiparticle Dispersion

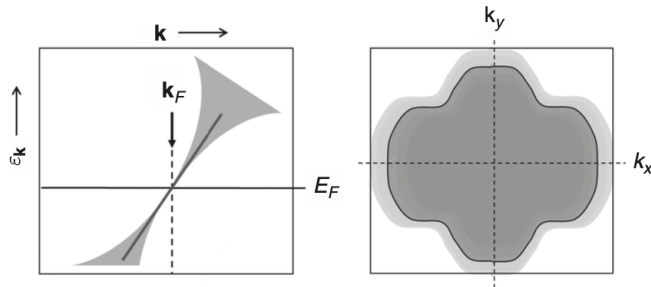


Figure 7.3. Schematic illustration of the width and dispersion of the quasi-particle structure in the spectral function Eq. (7.22) for addition and removal of electrons close to the Fermi level. In the left panel, the solid line indicates the quasi-particle energy $\epsilon_{\mathbf{k}}$ as a function of $\mathbf{k}$ near the Fermi momentum $\mathbf{k}_F$ for a single direction in $\mathbf{k}$ -space. The shaded area represents the width of $A_{\mathbf{k}}(\omega)$ around $\omega = \epsilon_{\mathbf{k}}$, given by $\text{Im } \Sigma_{\mathbf{k}}(\epsilon_{\mathbf{k}}) \propto (\epsilon_{\mathbf{k}} - E_F)^2$ in Eq. (7.24); so long as the width is small on the energy scale of interest, there is a peak in the spectral function that can be identified as a quasi-particle. Because the lifetime must go to infinity for states at the Fermi energy, the energies are well-defined and their variation with $\mathbf{k}$ defines the Fermi surface, depicted on the right. The shaded regions indicate the spread of the occupation numbers $n_{\mathbf{k}}$ in an interacting system (see, e.g., Fig. 5.2 for the homogeneous electron gas). Still, there is a discontinuity and hence a sharp surface,

Section 7.6: Quasi-particle Equations

- ▶ Goal: Find an effective one-particle Schrödinger-like equation for quasi-particles.
- ▶ Start from the Dyson equation and view the self-energy as a frequency-dependent potential.
- ▶ Expand the eigenfunctions in a single-particle basis:

$$\tilde{\psi}_{nk}(r, z) = \sum_m b_{mk}(z) \psi_{mk}(r).$$

(Eq. 7.34)

Effective Schrödinger Equation

- ▶ In matrix form:

$$\left[\varepsilon_k^0 + \Sigma_k^{\text{cont}}(z) \right] b_k(z) = \tilde{\varepsilon}_k(z) b_k(z),$$

(Eq. 7.35)

- ▶ The analytic continuation of the Green's function is given by:

$$G_k^{\text{cont}}(z) = \frac{b_k(z) \bar{b}_k(z)}{z - \tilde{\varepsilon}_k(z)}.$$

(Eq. 7.36)

Quasi-particle Pole and Renormalization

- ▶ Near the quasi-particle pole $z \approx \tilde{\epsilon}_k$:

$$G_k^{QP}(z) = \frac{Z_k b_k(\tilde{\epsilon}_k) \bar{b}_k(\tilde{\epsilon}_k)}{z - \tilde{\epsilon}_k},$$

(Eq. 7.37)

- ▶ Z_k is the renormalization factor, accounting for the strength (weight) of the quasi-particle.

Simplified Quasi-particle Equation

- ▶ Neglecting the imaginary part of the self-energy yields a Hermitian equation:

$$\left[\varepsilon_k^0 + \text{Re } \Sigma_k(\varepsilon_k) \right] b_k(\varepsilon_k) = \varepsilon_k b_k(\varepsilon_k).$$

(Eq. 7.39)

- ▶ This equation is most commonly used to calculate the quasi-particle energies (band structure) near the Fermi level.

Section 7.7: Coupled Dyson Equations

- ▶ Often, the self-energy (or polarizability) is split into two parts:

$$\Sigma = \Sigma_1 + \Sigma_2.$$

- ▶ Introduce an intermediate Green's function:

$$G_1(\omega) = G_0(\omega) + G_0(\omega) \Sigma_1(\omega) G_1(\omega),$$

(Eq. 7.40)

- ▶ Then, the full Green's function is:

$$G(\omega) = G_1(\omega) + G_1(\omega) \Sigma_2(\omega) G(\omega).$$

(Eq. 7.41)

Applications and Self-Consistency

- ▶ The decomposition strategy can also be applied to screening:

$$W_1(\omega) = v_c + v_c P_1(\omega) W_1(\omega), \quad W(\omega) = W_1(\omega) + W_1(\omega) P_2(\omega) W(\omega).$$

(Eqs. 7.42 & 7.43)

- ▶ Practical applications include:
 - ▶ Separating static and dynamical contributions (e.g., in GW, COHSEX, or G_0W_0).
 - ▶ Embedding approaches such as DMFT, where localized orbitals are treated differently.
- ▶ Self-consistency is often required, as one part may depend on the full solution.

Summary

- ▶ We formulated the Green's function in a reduced Hilbert space and derived the Dyson equation.
- ▶ The self-energy Σ encapsulates the influence of the rest of the system.
- ▶ A simple example (a discrete state coupled to a continuum) shows how to derive the effective Green's function.
- ▶ In interacting systems, the spectral function splits into a coherent quasi-particle peak and satellites.
- ▶ Quasi-particle equations are derived from the Dyson equation; the renormalization factor Z_k plays a key role.
- ▶ Coupled Dyson equations allow for efficient decomposition of complex many-body effects.